

THIRD-PARTY IMS APPLICATION SERVER

B2BUA Number Translation & Routing POC

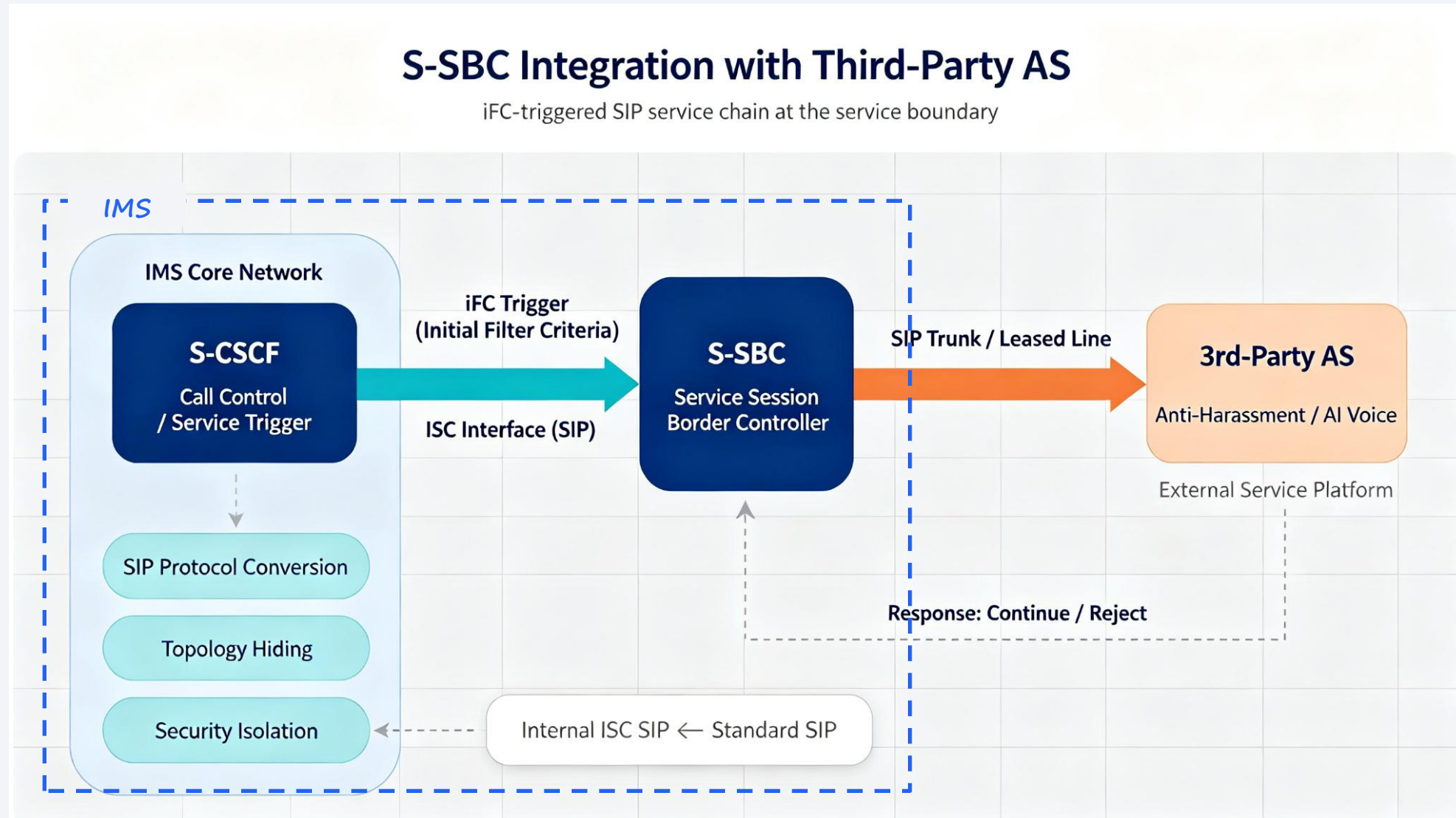
Signalling-only SIP B2BUA - Declarative YAML routing - Carrier-grade observability

Version 0.5.0 · SIP / RFC 3261 · Signalling Only

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B2BUA outside the carrier network, signalling only



Three-process isolation in Docker Compose

DOCKER HOST (compose)

Container: s-sbc-mock

s_sbc_mock

`python -m s_sbc_mock.main`

UDP 15060 - UAC side

UDP 15061 - UAS side

Simulates S-CSCF trigger

Simulates core network

Same SIP stack as AS

Configurable via .env

Container: as

as_app

`python -m as_app.main`

UDP 5060 - SIP trunk

TCP 8080 - internal API

B2BUU core

Routing engine

Number translation

Structured logging

Health check :8080

Container: console

console_app

`python -m console_app.main`

TCP 8000 - Web UI

HTTP proxy to :8080

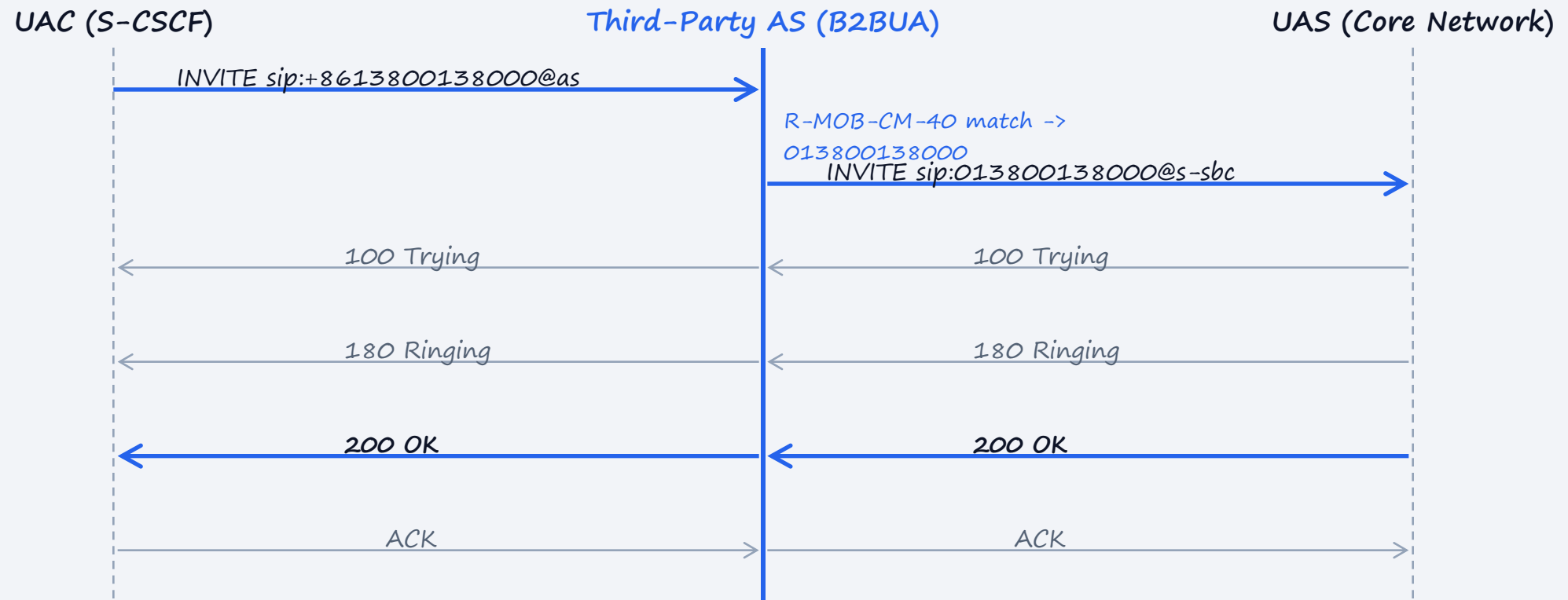
Dark ops dashboard

Real-time message flow

SVG topology view

Stats panel

Full call flow: INVITE relayed after number translation



SIP headers and SDP body pass through verbatim; only the Request-URI number is rewritten. The Call-ID links both legs for end-to-end tracing.

Declarative YAML rules: data is policy, hot-reloaded

Rule example (config/routing_rules.yaml)

```
# Mobile number translation
- rule_id: R-MOB-CM-40
  priority: 100
  match:
    called_prefixes:
      ["+86134", "+86135", ...]
  number_format: e164
  action:
    kind: route
  translate:
    to_format: national
  strip_prefix: "+86"
  prepend: "0"
  next_hops: [s-sbc-primary, s-sbc-failover]
```

Rule set overview

17

Routing rules

6

Next hops (primary + backup)

4

Number formats

Priority match - rules evaluated by priority ascending, first match wins

Exact + prefix - called_numbers exact or called_prefixes prefix

Format-aware - filter by e164 / national / short_code / international

Hot reload - 1s poll detects file changes; bad config falls back to old rules

Failover - next_hops list tried in order on failure

Four number formats interconverted: E.164, national, short, intl

Scenario	Inbound	Outbound	Rule / logic
Mobile	+8613800138000	013800138000	R-MOB-CM-40 - strip +86, prepend 0
Shanghai fixed	+862161800001	02161800001	R-FIX-SH-60 - strip +86, prepend 0
PBX extension	6001	+86216186001	R-PBX-30 - prefix +8621618
International	0012125550100	+12125550100	R-INTL-80 - strip 00, prepend +
Emergency	110	110	R-EMG-01 - pass through unchanged
Blocked service	+861681234567	603 Decline	R-BLOCK-90 - policy rejection

Four error branches with explicit SIP responses

404

No matching rule

Callee matches no enabled rule
Error: AS-ROUTE-001
e.g. +9991234567

603

Policy rejection

Rule action = reject (blocked services)
Error: AS-ROUTE-002
e.g. +861681234567

CANCEL

Caller cancels

Caller sends CANCEL before answer
Response: 487 Request Terminated
Release upstream leg

403

Unknown peer

Source not in whitelist
Error: AS-PEER-001

Unified error code system

All internal errors use the **AS-XXX-NNN** format, mapped to SIP status codes in errors.py. Every error has structured logging fields and an acceptance test.

Next-hop failover: when the primary next hop does not respond within 3 seconds, the backup is tried automatically - transparent to the caller.

Carrier-grade observability: structured logs, Call-ID tracing

Structured logging & end-to-end tracing

```
{  
  "timestamp": "2026-09-16T10:23:41Z",  
  "level": "INFO",  
  "call_id": "call-001@office.example",  
  "direction": "inbound",  
  "peer": "127.0.0.1:15060",  
  "rule_id": "R-MOB-CM-40",  
  "event": "call_relayed"  
}
```

Every log line carries the Call-ID, linking both legs of a call. Built-in counters cover total calls, disposition breakdown, error code distribution, rule hit distribution, and peer status.

Ops console (separate process)

[Real-time message flow](#) - direction color-coded, Call-ID filter, payload expansion

[Rule hit highlight](#) - which rule each call matched, at a glance

[Stats panel](#) - call volume, disposition, rule hit bar charts

[SVG topology](#) - real-time call path from S-SBC to next hop

[Dark ops theme](#) - pure HTML/CSS/JS, zero frontend dependencies

[WebSocket push](#) - 1s poll of TraceRecorder for incremental events

Six architecture decision records lock the tech stack

Tech stack (locked)

<i>Language</i>	<i>Python 3.10</i>
<i>SIP stack</i>	<i>sippy 2.4.2 (RFC 3261)</i>
<i>Web</i>	<i>FastAPI + uvicorn</i>
<i>Deps</i>	<i>uv + uv.lock</i>
<i>Quality</i>	<i>ruff - mypy - pytest</i>
<i>Deploy</i>	<i>docker compose</i>
<i>CICD</i>	<i>github action</i>
<i>License</i>	<i>Apache-2.0</i>

Architecture decision records

[ADR-0001](#) - sippy chosen as SIP stack and B2BUA framework

[ADR-0002](#) - console as separate process, internal API

[ADR-0003](#) - UDP only (POC scope)

[ADR-0004](#) - declarative YAML rules with hot reload

[ADR-0005](#) - Mock S-SBC on same SIP stack as AS

[ADR-0006](#) - signalling plane only, no media

All M0-M4 milestones complete, v0.5.0 ready



Known restrictions and items to be completed

RESTRICTIONS

Current POC limitations by design

- UDP transport only, no TCP or TLS
- Source IP whitelist only, no SIP Digest auth
- Single peer pair, no multi-S-SBC routing
- No persistence; calls lost on restart
- No CDR generation or billing interface
- No DoS protection, rate limiting, or CAC
- Local file logs only, no central collection
- Local YAML + .env config only

TODO

Roadmap to production-grade

- Add TCP and TLS transport (SIPS)
- SIP Digest authentication and TLS certs
- Multi-S-SBC peer group with smart failover
- Session persistence and restart recovery
- Per-call CDR generation and reporting
- Rate limiting, CAC, and black/white lists
- Centralized log collection and alerting
- Managed config service and secret management

THIRD-PARTY IMS / SIP APPLICATION SERVER

Signalling-only B2BUA, POC verified

Declarative routing rules – multi-format number translation – carrier-grade observability

v0.5.0

Current version

119

tests all passing

6 ADR

architecture decisions

Apache-2.0

Open source license

Demo - Call Trace

● ok

VER 0.5.0

UPTIME 414s

CALLS 1

PEERS 127.0.0.1:15060:trunk:reachable, s-sbc-primary:127.0.0.1:15061:reachable

live

Call Trace

Rules

Configuration

Statistics

About

filter Call-ID ...

c46ad4d33217d3a982ccc57e99c1b7c5

10 events · R-MOB-CM-40

c46ad4d33217d3a982ccc57e99c1b7c5

S-SBC

R-MOB-CM-40

AS

NH

← INVITE invite received from the trunk 20:03:13 127.0.0.1:15060

leg:

trunk

called_number:

+8613800138000

• decision route:

China Mobile subscribers, E.164 in and national format out R-MOB-CM-40

20:03:13 -

→ INVITE invite originated towards the next hop 20:03:13 127.0.0.1:15061

leg:

next_hop

called_number:

013800138000

next_hop:

s-sbc-primary

Demo cont - Rules

● ok

VER 0.5.0

UPTIME 657s

CALLS 1

PEERS 127.0.0.1:15060:trunk:reachable, s-sbc-primary:127.0.0.1:15061:reachable

live

Call Trace

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Next Hops

Routing Rules

NAME	ADDRESS	PORT	PRI	DESCRIPTION
s-sbc-primary	127.0.0.1	15061	1	Operator Service-SBC, primary trunk (mock in this POC)
s-sbc-failover	127.0.0.1	15062	2	Operator Service-SBC, second trunk used on failover
office-pbx-primary	127.0.0.1	15063	1	Office PBX, internal extensions
office-pbx-secondary	127.0.0.1	15064	2	Office PBX, standby node
intl-gateway-primary	127.0.0.1	15065	1	International gateway, primary
intl-gateway-secondary	127.0.0.1	15066	2	International gateway, standby

RULE ID	PRI	ENABLED	DESCRIPTION	MATCH	ACTION
R-EMG-01	10	on	Emergency short codes are routed first and never translated	exact:110,119,120,122	route hops:s-sbc-primary,s-sbc-failover
R-SVC-10	30	on	Operator and public service short codes	exact:10086,10010,10000,95588,95566,95598,12306	route hops:s-sbc-primary,s-sbc-failover
R-SHORT-20	35	on	Toll-free and shared-cost service numbers	prefix:400,800	route hops:s-sbc-primary,s-sbc-failover
R-PBX-30	40	on	Internal office extensions are expanded to the E.164 office range	prefix:60,61,62,63,64,65,66,67,68,69 fmt:short_code	route hops:office-pbx-primary,office-pbx-secondary ++8621618
R-MOB-CM-40	100	on	China Mobile subscribers, E.164 in and national format out	prefix:+86134,+86135,+86136,+86137,+86138,+86139,+86150,+86151,+86152,+86157,+86158,+86159,+86182,+86183,+86184,+86187,+86188,+86198 fmt:e164	route hops:s-sbc-primary,s-sbc-failover -->+86+0
R-MOB-CU-41	101	on	China Unicom subscribers, E.164 in and national format out	prefix:+86130,+86131,+86132,+86155,+86156,+86166,+86167,+86176,+86185,+86186 fmt:e164	route hops:s-sbc-primary,s-sbc-failover -->+86+0
R-MOB-CT-42	102	on	China Telecom subscribers, E.164 in and national format out	prefix:+86133,+86149,+86153,+86162,+86173,+86177,+86180,+86181,+86189,+86190,+86191,+86193,+86199 fmt:e164	route hops:s-sbc-primary,s-sbc-failover -->+86+0

Demo cont - Rules

● ok

VER 0.5.0

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